# p8106_hw2_sc5826

Shivalika Chavan

2026-03-02

## Data Cleaning & Splitting into Train/Test

```
college = read_csv("College.csv") |>
  janitor::clean_names()
```

```
## Rows: 565 Columns: 18
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): College
## dbl (17): Apps, Accept, Enroll, Top10perc, Top25perc, F.Undergrad, P.Undergr...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
set.seed(1234)
indexTrain = createDataPartition(y = college$outstate, p = 0.8, list = FALSE)
college_train = college[indexTrain, ]
college_test = college[-indexTrain, ]
```

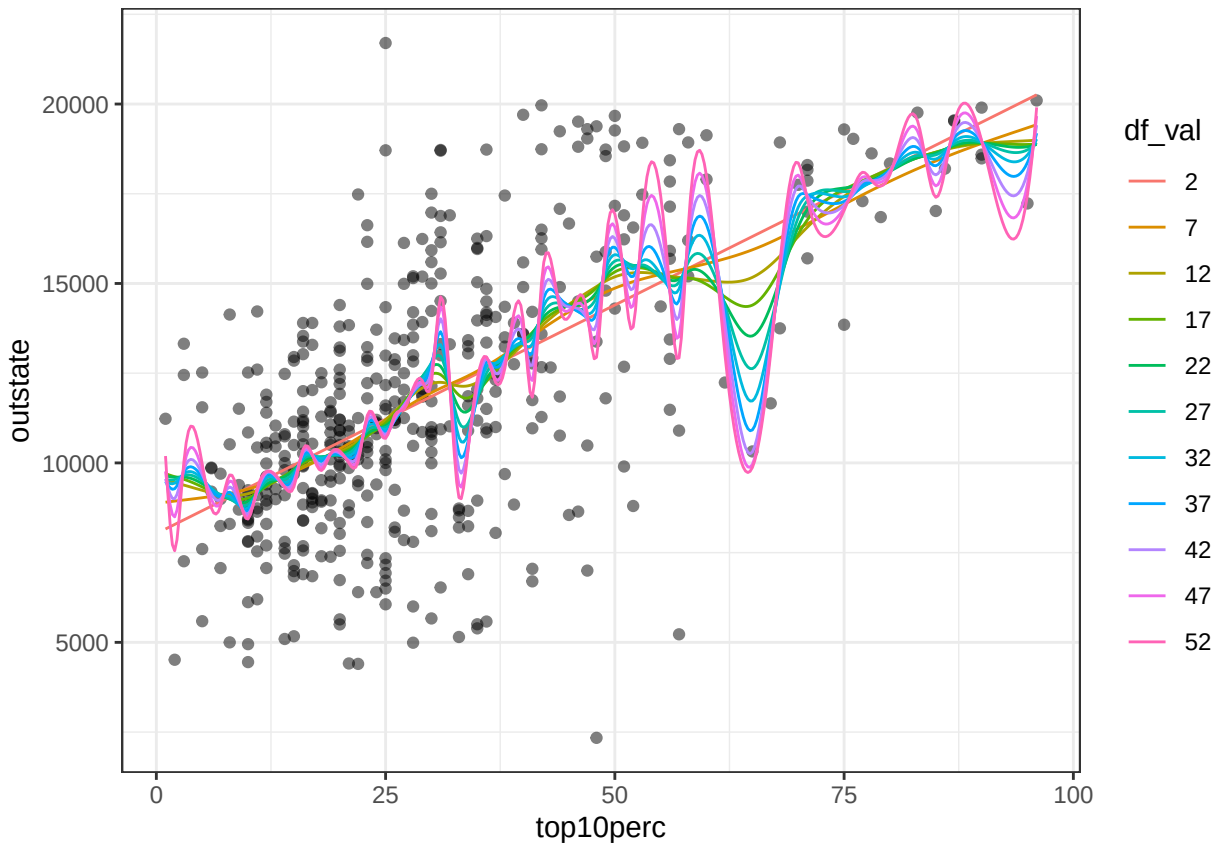
## A - Smoothing Spline

Fit smoothing spline models to predict out-of-state tuition (Outstate) using Top10perc as the only predictor, across a range of degrees of freedom. Plot the fitted curve for each degree of freedom. Describe the patterns you observe as the degrees of freedom change. Choose an appropriate degree of freedom for the model and plot this optimal fit. State the criteria you used to select the degree of freedom.

```
set.seed(1234)
top10perc_grid = seq(min(college_train$top10perc), max(college_train$top10perc), by = 0.25)

spline_df_fits =
  tibble(df_val = seq(from = 2, to = 52, by = 5)) |>
  mutate(
    spline_model = map(df_val, \(df) smooth.spline(x = college_train$top10perc, y = college_train$outsta
    predictions = map(spline_model, \(model) predict(model, x = top10perc_grid) |> as.data.frame())
  ) |>
  unnest(predictions) |>
  mutate(df_val = as.factor(df_val))

ggplot(college_train, aes(x = top10perc, y = outstate)) +
  geom_point(alpha = 0.5) +
  geom_line(data = spline_df_fits, aes(x = x, y = y, color = df_val)) +
  theme_bw()
```




As the degrees of freedom increase, the smoothing spline line deviates more from a straight line. At the lowest degrees of freedom (`df = 2`), the model has a heavier penalty for roughness, resulting in a nearly straight line. At the other end (`df = 52`), there is a lighter penalty for roughness, and the spline adjusts more to individual points. At higher degrees of freedom, there is a risk of overfitting the training data.

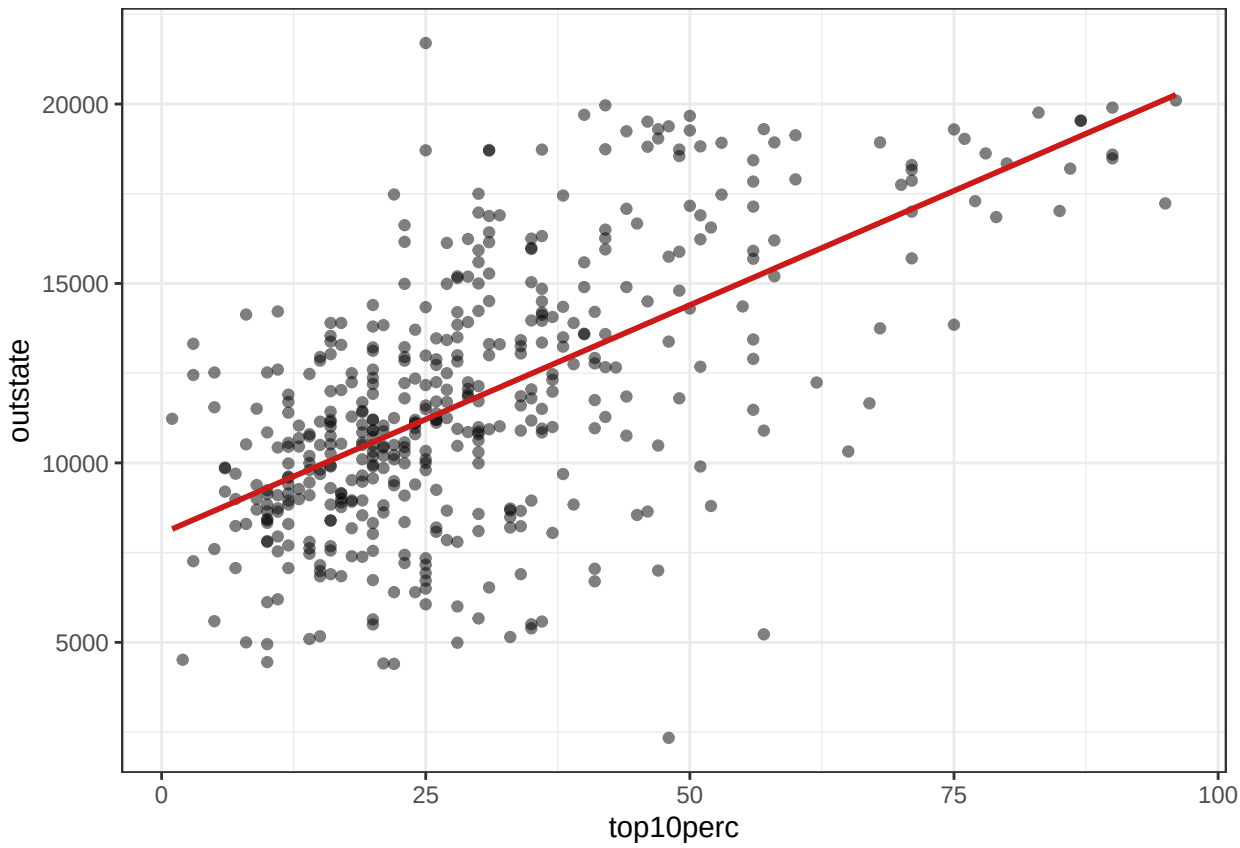
To select the optimal degrees of freedom, I used the default smoothing spline function, without specifying the degrees of freedome. This calculates and minimizes the default Generalized Cross-Validation (GCV) error to select the tuning parameter ($\lambda$).

```
best_smoothing_spline = smooth.spline(x = college_train$top10perc, y = college_train$outstate)

optimal_df = best_smoothing_spline$df

optimal_pred_df = predict(best_smoothing_spline, x = top10perc_grid) |>
  as.data.frame()

ggplot(college_train, aes(x = top10perc, y = outstate)) +
  geom_point(alpha = 0.5) +
  geom_line(data = optimal_pred_df, aes(x = x, y = y), color = rgb(.8, .1, .1, 1), linewidth = 1) +
  theme_bw()
```




Based on the above criterion, the optimal degrees of freedom chosen for the smoothing spline model was ~2.